

V4.2 Toy Model 4C

Curvature from Varying Lorentzian Metric Candidate

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | Continuation after Toy Model 4B

Status: Toy Model 4C computes Christoffel symbols, Ricci tensor, Ricci scalar, and Einstein-tensor candidate from a controlled varying Lorentzian metric. It does not derive Einstein Field Equations, physical stress-energy, gravity, or the dark sector.

1. Purpose

Toy Model 4B produced a Lorentzian signature candidate by adding directed 1T causal ordering to a spatial metric candidate. Toy Model 4C asks the next question: if the Lorentzian metric varies across the graph, do the standard mathematical curvature objects appear?

The target is not yet gravity. The target is the tensor-calculus bridge: metric gradients -> Christoffel symbols -> Ricci tensor -> scalar curvature.

2. Important correction before interpretation

Curvature is not the same as Einstein dynamics. A metric can be curved without yet proving what source created it. In General Relativity the source relation is the Einstein Field Equation, which connects curvature to stress-energy. Toy Model 4C does not derive that equation.

Therefore, this report tests curvature extraction from a controlled metric field. It does not prove that relational density physically causes gravity.

3. Controlled metric used for the test

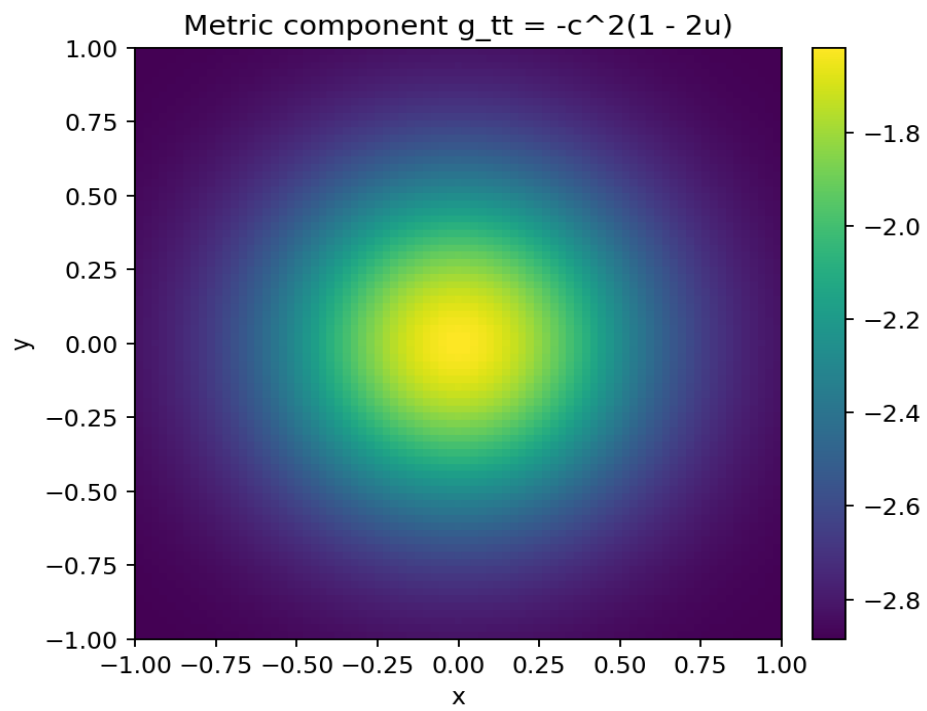
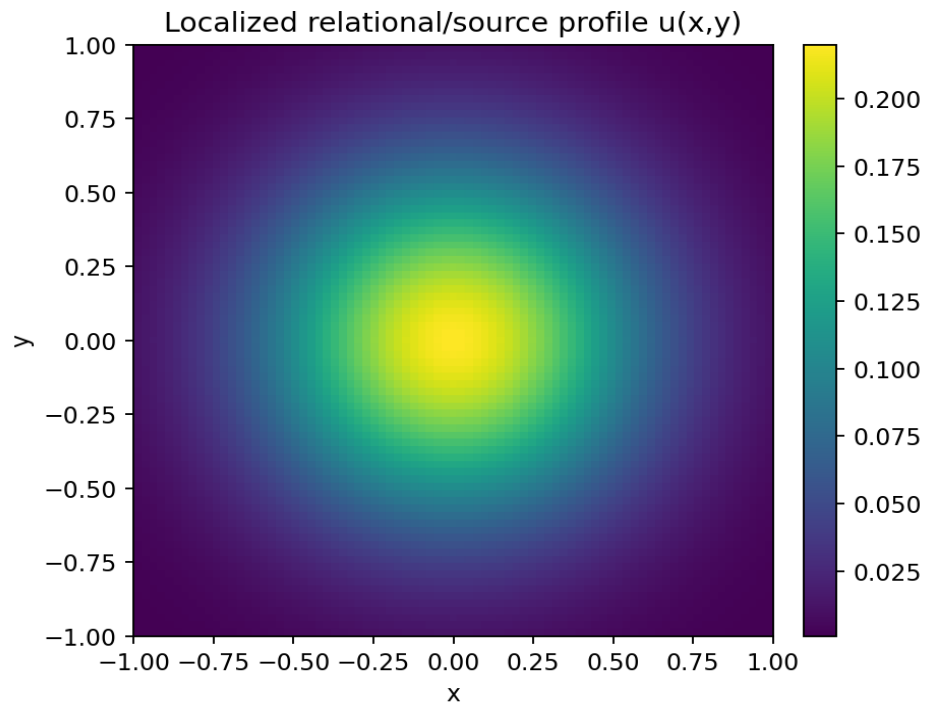
The test uses a static 1+2D Lorentzian metric with a localized source-like profile:

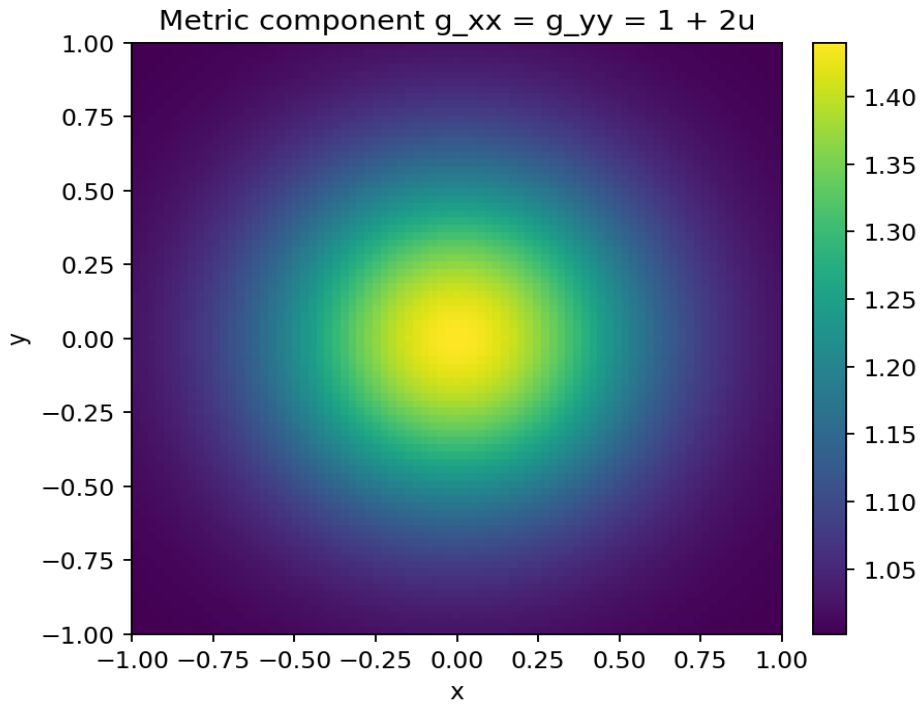
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u(x,y) = epsilon exp[-(x^2 + y^2)/(2 rho^2)]  
  
g_tt = -c^2(1 - 2u)  
g_xx = 1 + 2u  
g_yy = 1 + 2u  
g_tx = g_ty = g_xy = 0
```

Meaning of symbols:

- $u(x,y)$ is a localized relational/source profile.
- ϵ controls source strength.
- ρ controls source spread.
- g_{tt} is the time-time metric component.
- g_{xx} and g_{yy} are spatial metric components.
- c is the causal conversion factor from 4B; it is inserted, not derived.

The metric is weak-field-inspired, but it is not claimed to be an exact solution of Einstein's equations.





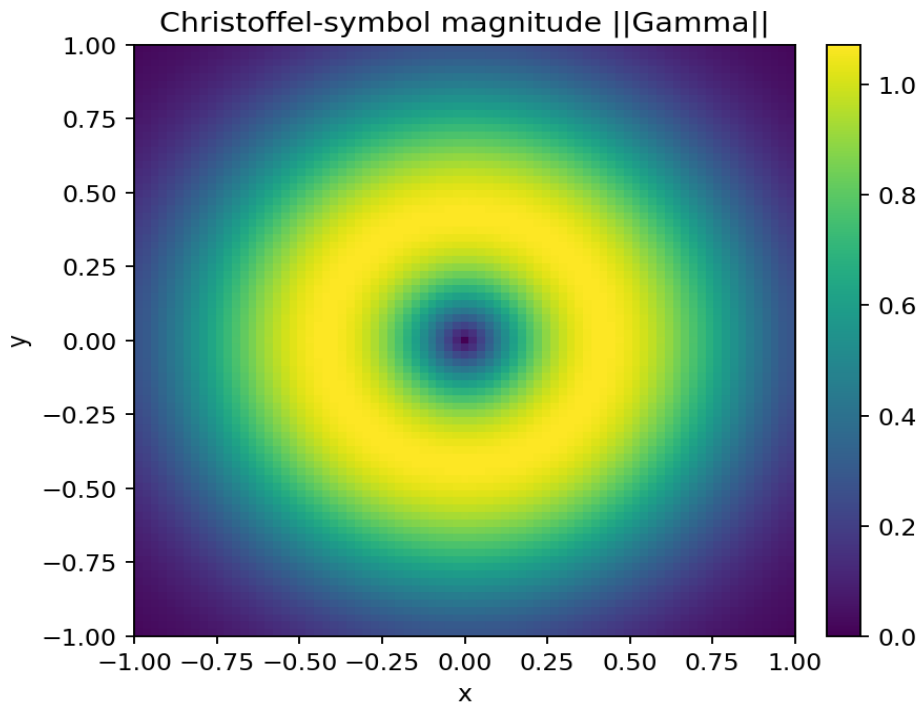
4. Christoffel symbols: the connection

The Christoffel symbols are computed from the metric and its gradients:

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} \left(\partial_{\mu} g_{\nu\sigma} + \partial_{\nu} g_{\mu\sigma} - \partial_{\sigma} g_{\mu\nu} \right)$$

Plain meaning: the Christoffel symbols tell how local coordinate directions change from point to point. They are not curvature by themselves. They are the connection used to define geodesics and curvature.

If the metric is constant, the Christoffel symbols vanish. If the metric varies, they become nonzero.

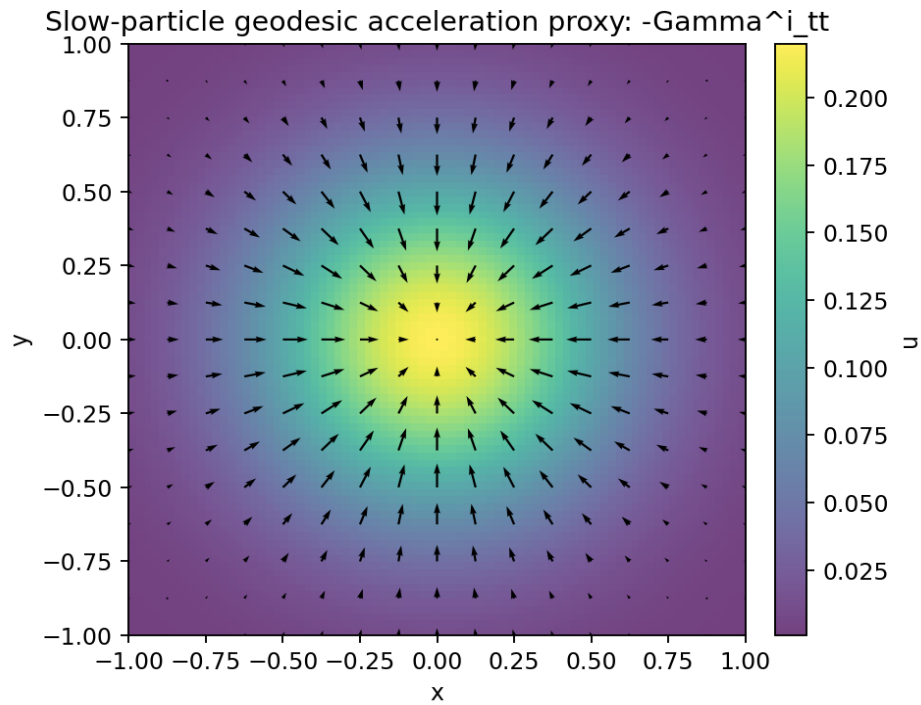


5. Geodesic acceleration proxy

For a slow particle in a static metric, a simple acceleration proxy is:

$$a^i \sim -\Gamma^i_{tt}$$

Meaning: the spatial acceleration is estimated from the Christoffel components that connect time-time curvature to spatial motion. This is only a weak, low-velocity diagnostic, not a full geodesic integration.



Geodesic diagnostic	Value
Max acceleration proxy magnitude	7.324246e-01
Fraction of tested vectors pointing inward	1.000000
Christoffel lower-index symmetry error	0.000000e+00
Metric compatibility max error	2.220446e-16

6. Ricci tensor and Ricci scalar

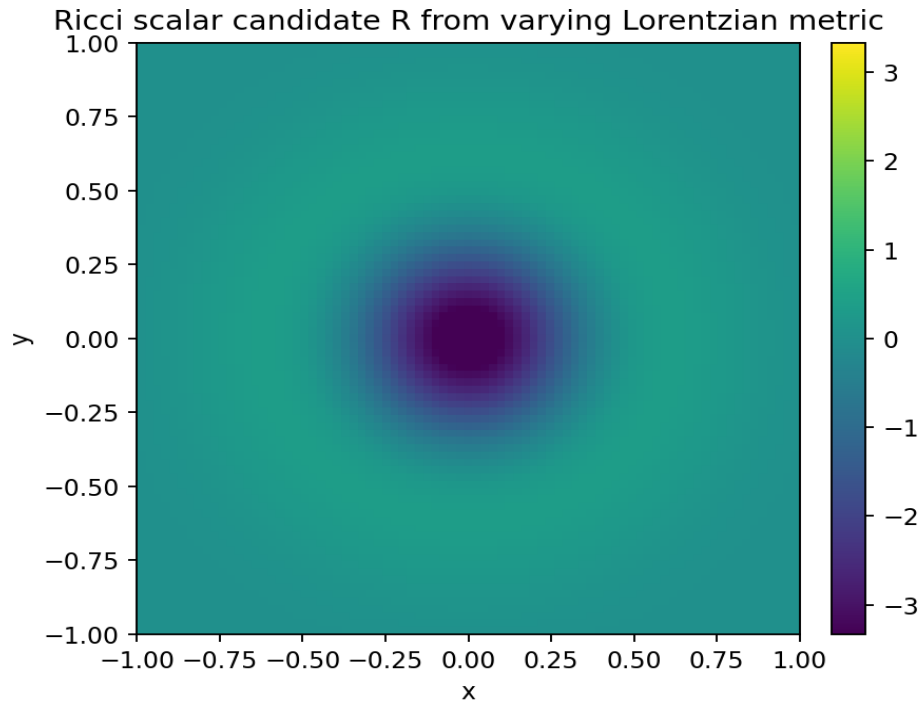
The Ricci tensor is computed from derivatives and products of Christoffel symbols:

$$\begin{aligned}
 R_{\mu\nu} = & \text{partial}_{\lambda} \Gamma^{\lambda}_{\mu\nu} \\
 & - \text{partial}_{\nu} \Gamma^{\lambda}_{\mu\lambda} \\
 & + \Gamma^{\lambda}_{\mu\nu} \Gamma^{\rho}_{\lambda\rho} \\
 & - \Gamma^{\rho}_{\mu\lambda} \Gamma^{\lambda}_{\nu\rho}
 \end{aligned}$$

The Ricci scalar is the contraction of the Ricci tensor with the inverse metric:

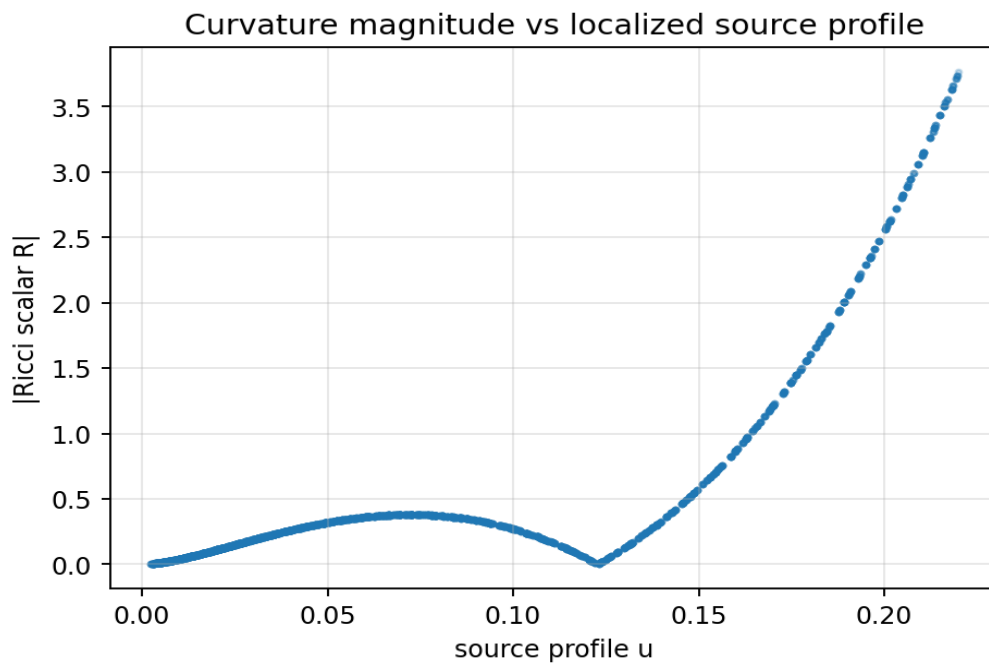
$$R = g^{\mu\nu} R_{\mu\nu}$$

Plain meaning: R is one compact scalar measure of curvature at each point. It cannot describe all curvature information, but it is useful for locating curvature concentration.



7. Curvature-source relation in the controlled toy model

The curvature is concentrated around the localized profile. The plot below compares source strength u with curvature magnitude $|R|$ across interior grid points.



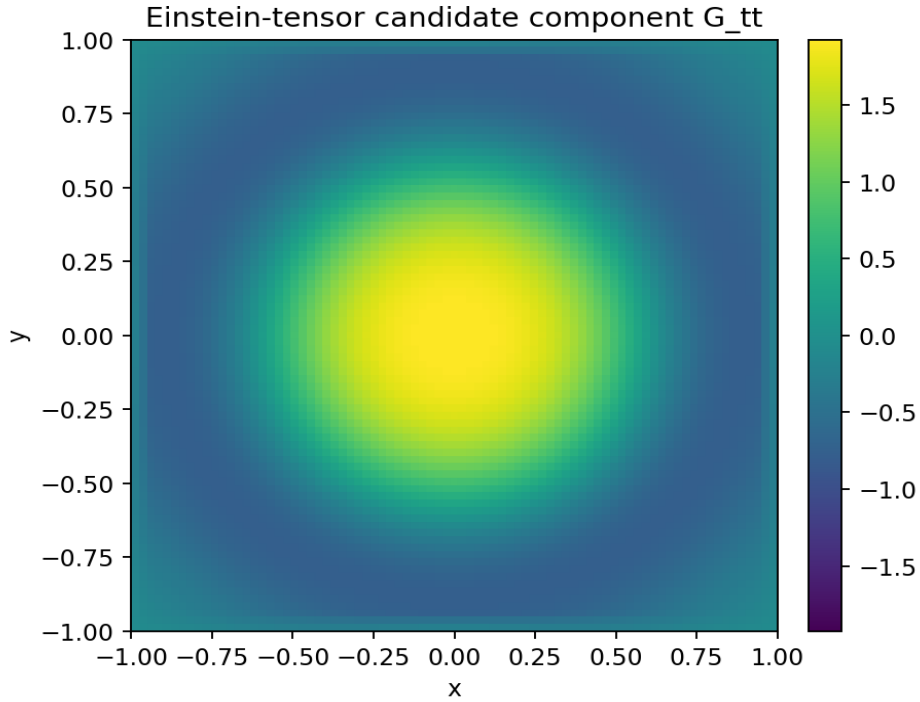
Curvature diagnostic	Value
Ricci scalar at center	-3.763266e+00
Minimum interior Ricci scalar	-3.763266e+00
Maximum interior Ricci scalar	3.805580e-01
Max interior $ R $	3.763266e+00
Mean edge $ R $ reference	5.986847e-02
$ R_{\text{center}} $ / edge mean $ R $	6.285890e+01
Correlation $ R $ vs source u	0.765999
Correlation R vs source u	-0.740887

8. Einstein-tensor candidate

After Ricci curvature is available, the Einstein-tensor candidate can be computed algebraically:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

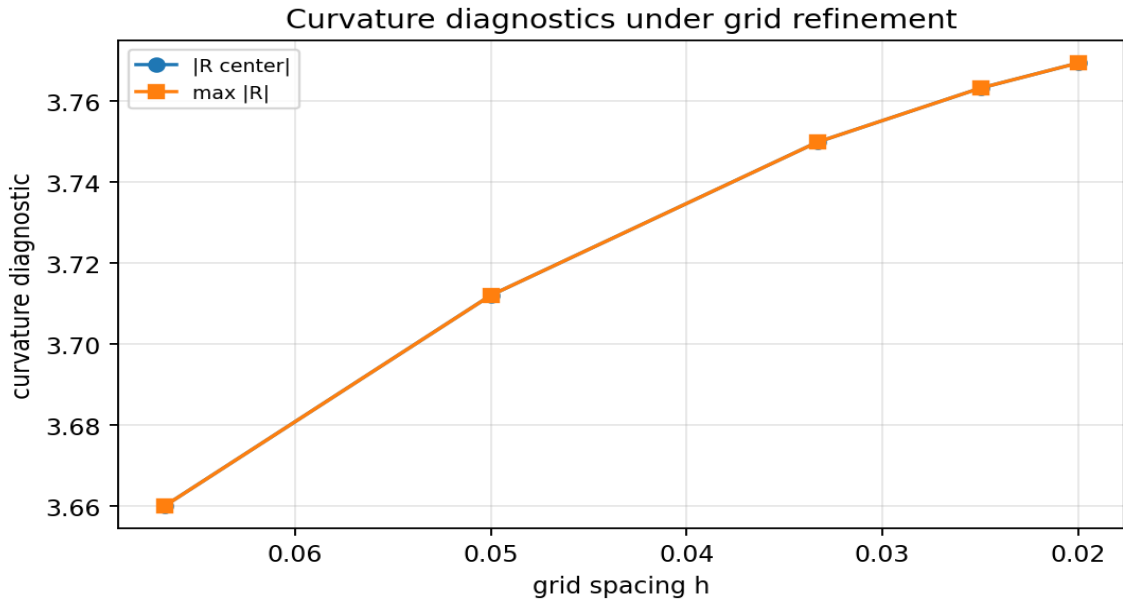
This is the left side of Einstein's Field Equation. In Toy Model 4C it is calculated only as a geometric object. The report does not claim to have produced the physical stress-energy tensor $T_{\mu\nu}$ or the coupling constant κ .



Einstein-candidate diagnostic	Value
G_{tt} at center	1.945753e+00
Max interior $ G_{tt} $	1.945753e+00
Ricci symmetry error	9.159518e-04

9. Signature and refinement checks

The metric retained Lorentzian signature throughout the interior calculation region. Curvature diagnostics were checked under grid refinement.



N	h	R center	max $ R $	corr $ R /u$	Lorentz sig.
31	0.06667	-3.660060e+00	3.660060e+00	0.759059	625/625
41	0.05000	-3.712102e+00	3.712102e+00	0.761453	1225/1225
61	0.03333	-3.749907e+00	3.749907e+00	0.766867	3025/3025

N	h	R center	max R	corr R /u	Lorentz sig.
81	0.02500	-3.763266e+00	3.763266e+00	0.765999	5329/5329
101	0.02000	-3.769472e+00	3.769472e+00	0.765654	8281/8281
Signature diagnostic				Value	
Interior metric points				5329	
Successful Lorentzian signatures				5329/5329	
Signature success ratio				1.000000	

10. Interpretation inside the V4.2 framework

Toy Model 4C supports this limited ladder:

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4B Lorentzian metric candidate
-> metric gradients
-> Christoffel connection
-> geodesic acceleration proxy
-> Ricci tensor
-> Ricci scalar
-> Einstein-tensor candidate

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It does not yet support this stronger ladder:

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relational density
-> physical stress-energy tensor
-> Einstein Field Equations
-> real gravity
-> dark matter / dark energy explanation

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11. What Toy Model 4C achieves

- It computes Christoffel symbols from a varying Lorentzian metric candidate.
- It verifies Christoffel lower-index symmetry.
- It verifies metric compatibility numerically.
- It computes Ricci tensor and Ricci scalar over the grid.
- It shows curvature concentration around a localized controlled source profile.
- It computes the Einstein-tensor candidate as a geometric object.
- It preserves Lorentzian signature across the tested interior region.

12. What Toy Model 4C does not achieve

- It does not derive Einstein Field Equations.
- It does not derive a physical stress-energy tensor.
- It does not derive the coupling constant between source and curvature.
- It does not prove that relational density physically causes curvature.
- It does not solve gravity.
- It does not explain dark matter or dark energy.
- It does not prove physical 3T or graph-spacetime ontology.

13. Close-out statement for 4C

Toy Model 4C successfully demonstrates that a varying Lorentzian metric candidate supports the standard differential-geometric curvature machinery. From the controlled metric field, finite-difference derivatives produce Christoffel symbols, Ricci tensor, Ricci scalar, and an Einstein-tensor candidate. Curvature localizes around the source-like profile and the geodesic acceleration proxy points inward. The strict guardrail remains: this is curvature extraction from an assumed metric profile, not a derivation of Einstein dynamics, physical stress-energy, gravity, or the dark sector.

14. Recommended next chapter

The next logical chapter is Toy Model 4D: Source-to-Curvature Calibration and Einstein-Tensor Consistency.

1. Define a candidate source density $S(t,x,y)$ from relational/coupling data.
2. Compare the computed Einstein-tensor component G_{tt} against S .

3. Test whether a proportional relation $G_{tt} \sim \kappa S$ can be calibrated.
4. Check whether other components remain physically interpretable.
5. Keep the guardrail: calibrating a toy source is not deriving physical stress-energy.

15. Rebind prompt for continuation

We are resuming after Toy Model 4C.

4A:

Relational graph distances $\rightarrow$ local Riemannian spatial metric candidate.

4B:

Add directed 1T causal order + signed interval:

$$ds^2 = -c^2 dt^2 + g_{ab} dx^a dx^b$$

This produced Lorentzian signature candidate.

4C:

Use varying Lorentzian metric:

$$u(x,y) = \epsilon \exp[-(x^2+y^2)/(2\rho^2)]$$

$$g_{tt} = -c^2(1 - 2u)$$

$$g_{xx} = g_{yy} = 1 + 2u$$

Compute:

$\Gamma^\lambda_{\mu\nu}$

Ricci tensor $R_{\mu\nu}$

Ricci scalar R

$$\text{Einstein-tensor candidate } G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

Result:

Curvature machinery works on the varying Lorentzian metric candidate.

Curvature localizes near the source-like profile.

A slow-particle geodesic acceleration proxy points inward.

Guardrail:

This is curvature extraction from an assumed metric profile.

It is not Einstein Field Equations, physical stress-energy, gravity, dark matter, dark energy, or proof of physical 3T.

Next recommended chapter:

Toy Model 4D - Source-to-Curvature Calibration and Einstein-Tensor Consistency.